

MAT11-1

Foundations

Logic, Real Analysis, Calculus, and Complex Numbers

The analytical and algebraic toolkit underpinning all subsequent mathematics.

Albert School — 38 hours

Contents

1	Logic foundations	2
1.1	Propositional logic	2
1.2	Analysis–synthesis reasoning	3
2	Mathematical notation and reading	3
3	Real numbers	3
4	Functions	4
4.1	Standard families	4
4.2	Trigonometric functions	4
5	Polynomials	4
6	Sequences	5
7	Limits and continuity	5
8	Derivatives	5
8.1	Extrema, convexity	6
8.2	Taylor approximations	6
9	Integration	6
10	Complex numbers	7

1 Logic foundations

1.1 Propositional logic

A *proposition* is a statement that is either true (T) or false (F). Compound propositions are built from simpler ones using connectives.

Definition 1 For propositions P and Q :

- *negation* $\neg P$ is true exactly when P is false;
- *conjunction* $P \wedge Q$ is true exactly when both P and Q are true;
- *disjunction* $P \vee Q$ is true exactly when at least one of P , Q is true;
- *implication* $P \Rightarrow Q$ is false only when P is true and Q is false;
- *equivalence* $P \Leftrightarrow Q$ is true exactly when P and Q have the same truth value.

A *truth table* lists the truth value of a compound proposition for every combination of truth values of its components.

P	Q	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Definition 2 The *contrapositive* of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$. An implication and its contrapositive are equivalent.

Proposition 3 Negations of compound statements:

$$\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q), \quad \neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q), \quad \neg(P \Rightarrow Q) \Leftrightarrow P \wedge (\neg Q).$$

1.2 Analysis–synthesis reasoning

To solve an existence/uniqueness problem by *analysis–synthesis*: in the *analysis* step, assume a solution exists and derive the constraints it must satisfy, narrowing the candidates; in the *synthesis* step, verify which candidates actually are solutions. Analysis establishes necessity; synthesis establishes sufficiency. Together they determine the full solution set.

2 Mathematical notation and reading

Definition 4 Set notation:

- $x \in A$: x is an element of A ;
- $A \subset B$: every element of A is an element of B ;
- $A \cup B = \{x : x \in A \vee x \in B\}$;
- $A \cap B = \{x : x \in A \wedge x \in B\}$.

The *summation* symbol abbreviates a sum:

$$\sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n.$$

The *universal quantifier* $\forall$ reads “for all” and the *existential quantifier* $\exists$ reads “there exists”. Reading a theorem statement means identifying its hypotheses (assumed) and its conclusion (asserted under those hypotheses), together with the quantifiers binding its variables.

3 Real numbers

The real numbers $\mathbb{R}$ form an *ordered field*: they are equipped with addition and multiplication satisfying the field axioms (associativity, commutativity, distributivity, additive identity 0 and inverses, multiplicative identity 1 and inverses of nonzero elements), together with a total order $\leq$ compatible with these operations (if $a \leq b$ then $a + c \leq b + c$; if $a \leq b$ and $0 \leq c$ then $ac \leq bc$).

Remark $\mathbb{R}$ is *complete*: the real line has no gaps. This is stated here and used informally; the formal completeness property is deferred to AD S3.

Definition 5 For $a \leq b$, intervals are $[a, b] = \{x : a \leq x \leq b\}$, $(a, b) = \{x : a < x < b\}$, and the half-open variants $[a, b)$, $(a, b]$.

Definition 6 The *absolute value* is $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$. The *distance* between x and y is $|x - y|$.

Theorem 7 (Triangle inequality) For all $x, y \in \mathbb{R}$,

$$|x + y| \leq |x| + |y|.$$

4 Functions

Definition 8 A function $f : A \rightarrow B$ assigns to each $x \in A$ a single value $f(x) \in B$. The set A is the *domain*. For $S \subset A$, the *image* is $f(S) = \{f(x) : x \in S\}$, and the *range* is $f(A)$. For $T \subset B$, the *preimage* is $f^{-1}(T) = \{x \in A : f(x) \in T\}$.

Definition 9 The *composition* of $f : A \rightarrow B$ and $g : B \rightarrow C$ is $(g \circ f)(x) = g(f(x))$.

Definition 10 A function $f : A \rightarrow B$ is:

- *injective* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$;
- *surjective* if $f(A) = B$;
- *bijective* if it is both injective and surjective.

A bijection f has an *inverse* $f^{-1} : B \rightarrow A$ with $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

4.1 Standard families

The *exponential* $\exp(x) = e^x$ and its inverse the *natural logarithm* $\ln$ satisfy

$$e^{x+y} = e^x e^y, \quad \ln(xy) = \ln x + \ln y, \quad \ln(e^x) = x \quad (x \in \mathbb{R}), \quad e^{\ln x} = x \quad (x > 0).$$

4.2 Trigonometric functions

On the *unit circle*, the angle θ (in radians) determines a point $(\cos \theta, \sin \theta)$. Define $\tan \theta = \sin \theta / \cos \theta$. Key identities:

$$\cos^2 \theta + \sin^2 \theta = 1,$$

$$\sin(a+b) = \sin a \cos b + \cos a \sin b, \quad \cos(a+b) = \cos a \cos b - \sin a \sin b.$$

Restricted to suitable domains, the trigonometric functions are bijective and admit *inverse trigonometric functions* $\arcsin$, $\arccos$, $\arctan$, with

$$\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2], \quad \arccos : [-1, 1] \rightarrow [0, \pi], \quad \arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2).$$

5 Polynomials

Definition 11 A *polynomial* is $P(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$; its *degree* is n . A *root* is a value r with $P(r) = 0$.

Theorem 12 (Factor theorem) r is a root of P if and only if $(x - r)$ divides $P(x)$. If $(x - r)^k$ divides $P(x)$ but $(x - r)^{k+1}$ does not, then r is a root of *multiplicity* k .

A polynomial of degree n has at most n roots, counted with multiplicity.

Theorem 13 (Vieta's formulas) If $P(x) = a_n x^n + \cdots + a_0$ has roots $r_1, \dots, r_n$ (with multiplicity), then

$$\sum_i r_i = -\frac{a_{n-1}}{a_n}, \quad \prod_i r_i = (-1)^n \frac{a_0}{a_n}.$$

In particular, for $x^2 + bx + c$ with roots r_1, r_2 : $r_1 + r_2 = -b$ and $r_1 r_2 = c$.

6 Sequences

Definition 14 An *arithmetic sequence* satisfies $u_{n+1} = u_n + d$, giving $u_n = u_0 + nd$. A *geometric sequence* satisfies $u_{n+1} = ru_n$, giving $u_n = u_0 r^n$.

Definition 15 A sequence (u_n) *converges* to L , written $\lim_{n \rightarrow \infty} u_n = L$, if for every $\varepsilon > 0$ there exists N such that $|u_n - L| < \varepsilon$ for all $n \geq N$.

Theorem 16 (Monotone convergence) A sequence that is increasing and bounded above converges (to its least upper bound). Likewise a decreasing sequence bounded below converges.

7 Limits and continuity

Definition 17 f is *continuous* at a if for every sequence (x_n) with $x_n \rightarrow a$ one has $f(x_n) \rightarrow f(a)$ (sequential criterion). Equivalently, for every $\varepsilon > 0$ there is $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \varepsilon$.

Polynomials, exp, ln, and the trigonometric functions are continuous on their domains; sums, products, quotients (where defined), and compositions of continuous functions are continuous.

Theorem 18 (Intermediate Value Theorem) If f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then $f(c) = y$ for some $c \in [a, b]$. In particular, if $f(a)$ and $f(b)$ have opposite signs, f has a root in (a, b) .

Theorem 19 (Extreme Value Theorem) A function continuous on a closed bounded interval $[a, b]$ attains a maximum and a minimum value on $[a, b]$.

8 Derivatives

Definition 20 f is *differentiable* at a with derivative

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

when this limit exists.

Proposition 21 (Differentiation rules)

$$(f + g)' = f' + g', \quad (fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Chain rule: $(g \circ f)'(x) = g'(f(x))f'(x)$. Inverse function derivative: $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$.

8.1 Extrema, convexity

Theorem 22 (Fermat's condition) If f has a local extremum at an interior point c and is differentiable there, then $f'(c) = 0$.

A point with $f'(c) = 0$ is a *critical point*. *First derivative test*: if f' changes from $+$ to $-$ at c , then c is a local maximum; from $-$ to $+$, a local minimum. *Second derivative test*: if $f'(c) = 0$ and $f''(c) > 0$, then c is a local minimum; if $f''(c) < 0$, a local maximum. A *local* extremum is best among nearby points; a *global* extremum is best over the whole domain.

Definition 23 f is *convex* on an interval if $f'' \geq 0$ there, and *concave* if $f'' \leq 0$. An *inflection point* is where convexity changes. Geometrically, a convex function lies above each of its tangent lines.

8.2 Taylor approximations

Definition 24 The *Taylor polynomial* of order n of f at a is

$$T_{n(x)} = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

In practice we use orders 1 and 2: near a , $f(x) \approx f(a) + f'(a)(x - a)$ (tangent line), and $f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$. Standard approximations for small x :

$$e^x \approx 1 + x, \quad \ln(1 - x) \approx -x, \quad \sin x \approx x.$$

9 Integration

Definition 25 An *antiderivative* of f is a function F with $F' = f$; the *indefinite integral* is $\int f(x) dx = F(x) + C$.

Theorem 26 (Fundamental Theorem of Calculus) If f is continuous on $[a, b]$ and F is an antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Moreover $x \rightarrow \int_a^x f(t) dt$ is an antiderivative of f , connecting differentiation and accumulation.

Proposition 27 (Integration by parts)

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, dx.$$

Proposition 28 (Substitution) With $u = g(x)$, $du = g'(x) \, dx$,

$$\int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

10 Complex numbers

Definition 29 A *complex number* in *algebraic form* is $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$. Its *modulus* is $|z| = \sqrt{a^2 + b^2}$ and an *argument* θ satisfies $a = |z| \cos \theta$, $b = |z| \sin \theta$. The *polar form* is $z = |z|(\cos \theta + i \sin \theta)$.

Theorem 30 (Euler's formula)

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad \text{so} \quad z = |z|e^{i\theta}.$$

Over $\mathbb{C}$, every polynomial of degree n has exactly n roots counted with multiplicity (Fundamental Theorem of Algebra). Equations such as $z^n = w$ are solved by writing w in polar form and taking the n roots $|w|^{1/n} e^{i(\theta+2\pi k)/n}$, $k = 0, \dots, n-1$.